



WASHINGTON STATE
UNIVERSITY

BROADBAND DATA FOR ALL

Who am I?

- Assistant Director of the Digital Equity and Broadband program at Washington State University
- Recovering IT person, no I will not help you fix your printer
- Curler, yes like with the stones and ice



How did we get here?

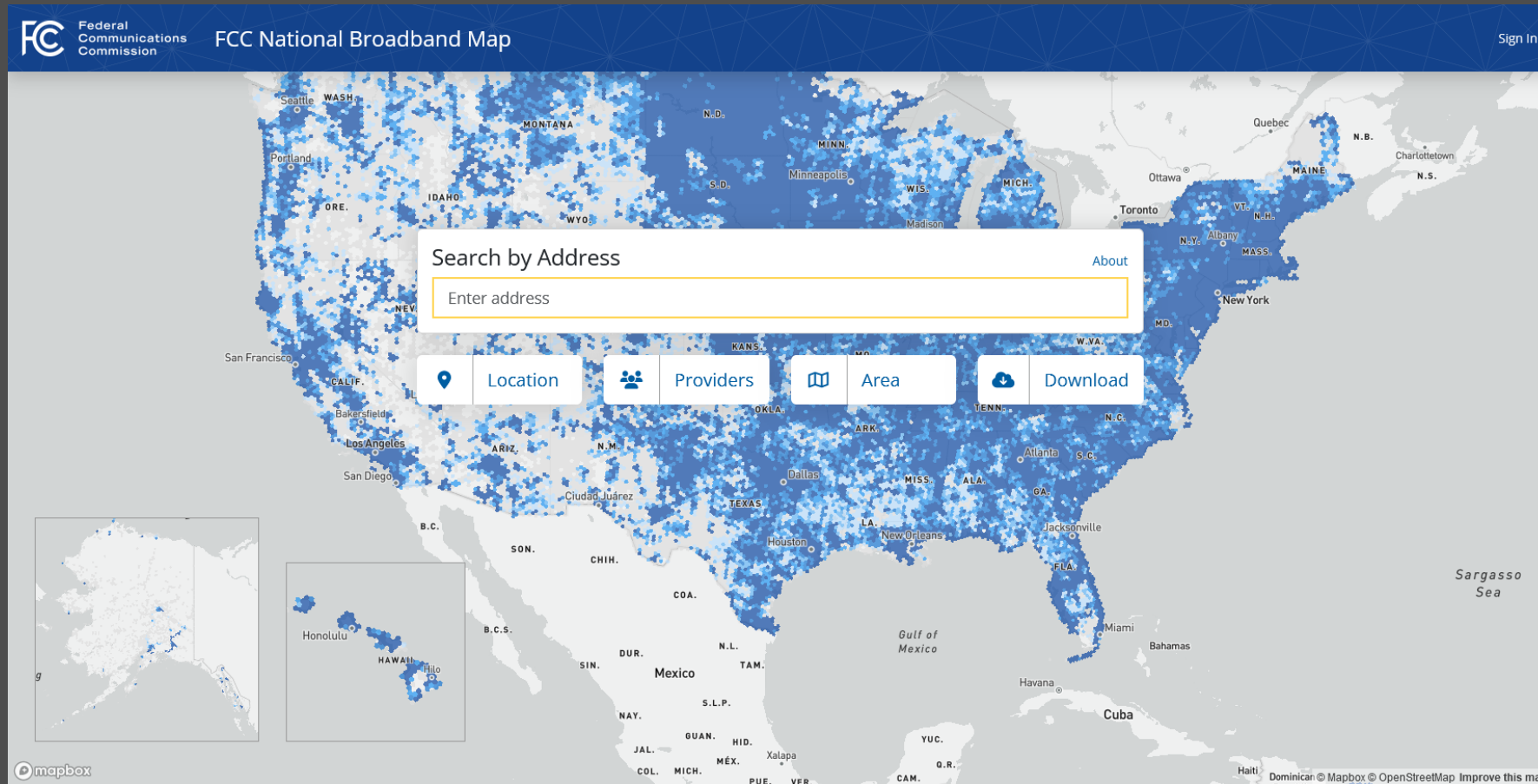


WSU

Broadband DATA Act

- Fully known as the “Broadband Deployment Accuracy and Technological Availability Act”
- Required the FCC to collect data about the availability of broadband services at the individual location level
- Collect that data at least twice a year
- Be compatible with common GIS software – Literally in the text of the act!!!
- It’s actually small enough to read: [Text - S.1822 - 116th Congress \(2019-2020\): Broadband DATA Act | Congress.gov | Library of Congress](#)

National Broadband Map



What is it?



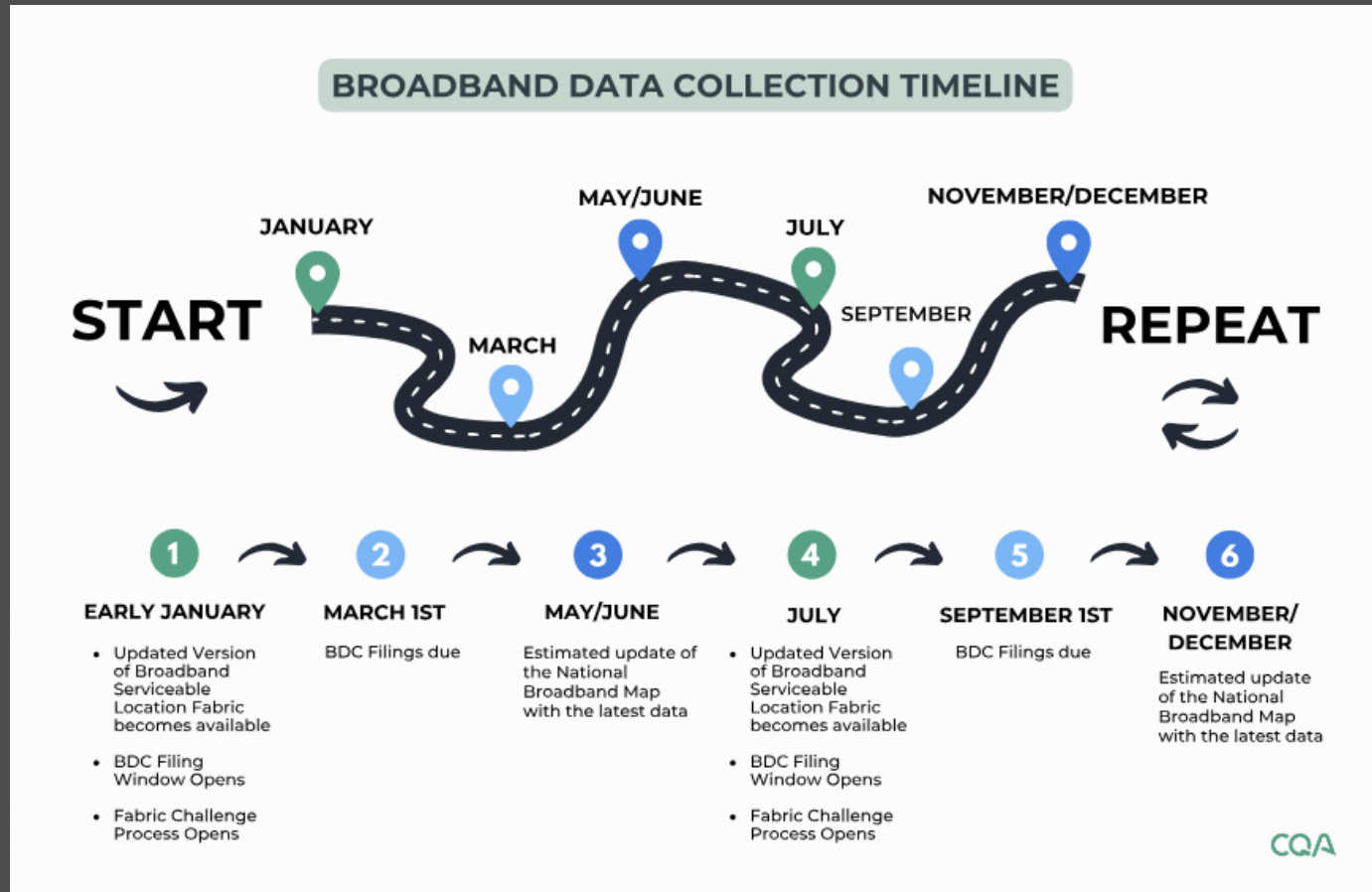
Hint: It's two things

WSU

It's two things

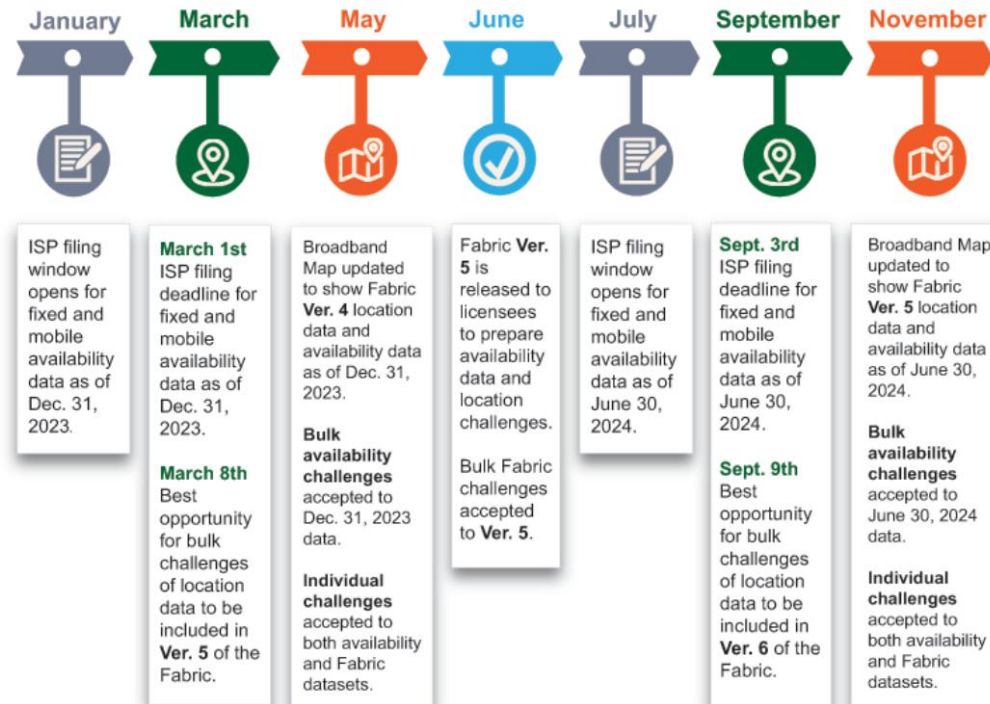
- Fabric – Released twice a year
 - It's a super fancy point layer
 - What is in it?
 - Unique Location ID
 - Address
 - Unit Count
 - Is it a BSL
 - Building/Land Use Code
 - Confidence Score
 - Census Block
 - H3 Res 9 Hexagon ID
 - Lat/Long
- Availability – Released every 2 weeks
 - It's an attribute table for Location IDs
 - What is in it?
 - Provider FRN and Unique Provider ID
 - Brand Name
 - Location ID
 - Technology providing service
 - Upload/Download Speed
 - Low latency
 - Residential/Business Service offering
 - USPS State Code
 - Census Block
 - H3 Res8 Hexagon ID

BDC Timeline



Key Dates 2024

National Broadband Map: 2024 Key Dates



What does it look like?



I heard it's really bad

WSU

What does it look like? (December 22 edition)



This is a BSL



So are these

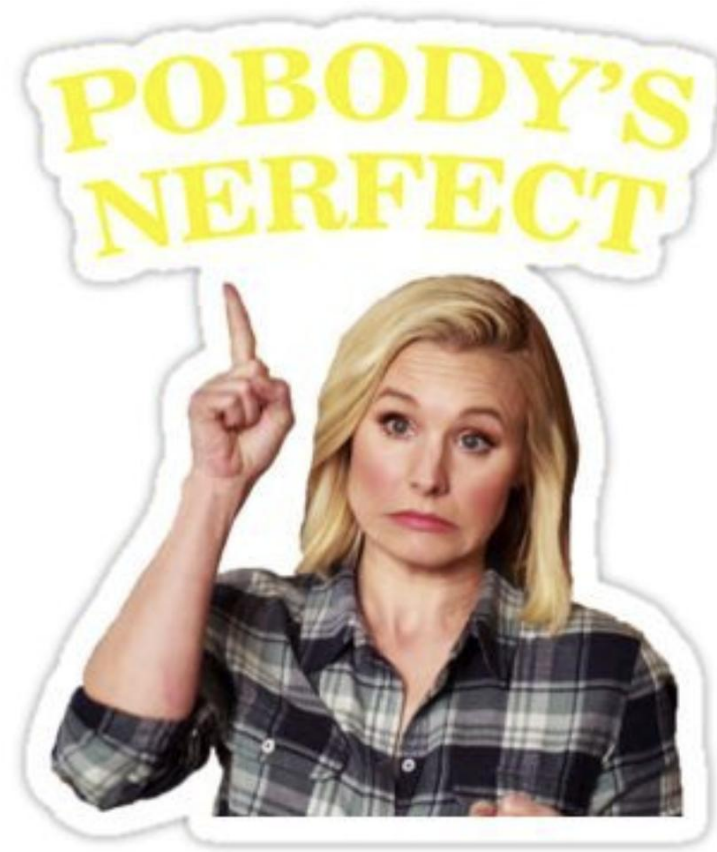


This too?



...





What does it look like? (December 22 edition)



What does it look like? (December 23 edition)



What disappeared?



But what about on the south side?



But what about on the south side?



This is quite tough.



I want the Fabric, how do I get it?



Hint: It takes a few steps

WSSU

How do I get it?

- Obtain an FCC Registration Number (FRN)
 - Mine is 0033104902
 - It isn't that hard to get one. It only takes about 15 minutes.

WSU

How do I get it?



- CostQuest Process
 - Fill out the form
 - Wait for them to get back to you (couple of days)
 - Work through your legal team (if you have one) to review the license
 - Download the CSV and process it
 - Do fun stuff!

WSU

How would I use it?



Hint: Pay attention to your license

WSSU

FCC Licenses



- For ISPs – Tier 2
 - Use for filings
 - Correct BDC data
- For Government Entities – Tier 2 Government
 - Use for challenges
 - Correct BDC data
 - Designated Entity may use fabric for planning
- For Others – Tier 4
 - Standard Agreement – Generating challenges
 - Research Agreement – Public policy/Academic research

WSU

NTIA Licenses



- Tier C
 - State Broadband Office
- Tier D
 - ISP or public entity participation
- Tier E
 - Public use for Challenge process

WSU

Process



- Download and enrich the CSVs
- Relate public data from the FCC to the licensed data points
- Do the analysis that you are looking to do
 - Remember your license limitations

WSU

Fabric data

- Fabric data removed due to licensing
- If you would like to discuss the fabric, please reach out to nick.pappin@wsu.edu

WSU

Availability data

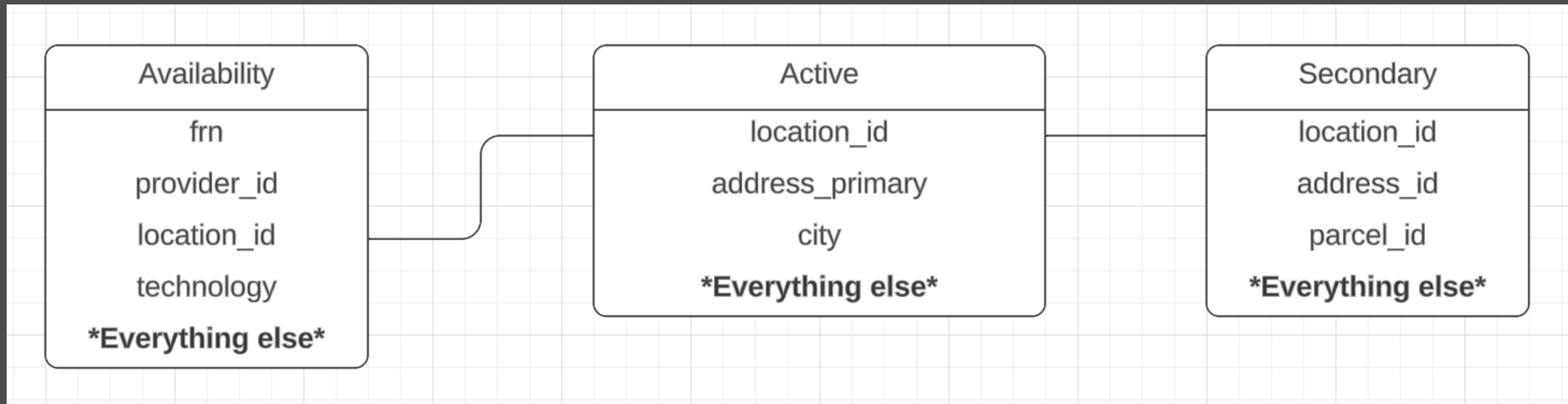
- 0003474327,130183,Sparklight,1145587130,40,1000,50,1,X,
WA,530039602002007,88288e6899ffff
- 0003474327,130183,Sparklight,1145592004,40,1000,50,1,X,
WA,530039606001008,88288e68b1ffff
- 0003474327,130183,Sparklight,1414826237,40,1000,50,1,X,
WA,530039603001035,88288ad251ffff
- 0005061544,130786,Ranier Connect,1330138602,40,1000,12,0,X,
WA,530539400051006,8828d5c337ffff
- 0005061544,130786,Ranier Connect,1330140101,40,1000,12,1,X,
WA,530539400132011,8828d5dagbffff
- 0005061544,130786,Ranier Connect,1330141873,40,1000,12,1,X,
WA,530539400062006,8828d5c26dffff

Availability Data

OBJECTID	501640
frn	29765211
provider_id	420173
brand_name	Ziply Fiber
location_id	1288946683
technology	50
max_advertised_download_speed	6000
max_advertised_upload_speed	6000
low_latency	1
business_residential_code	X
state_usps	WA
block_geoid	530750004001038
h3_res8_id	88288ac5cbffff

OBJECTID	584436
frn	25646373
provider_id	130235
brand_name	Charter Communications Inc
location_id	1288946625
technology	40
max_advertised_download_speed	1000
max_advertised_upload_speed	35
low_latency	1
business_residential_code	R
state_usps	WA
block_geoid	530750004001038
h3_res8_id	88288ac5cbffff

ERD Diagram



What can I do without doing that?

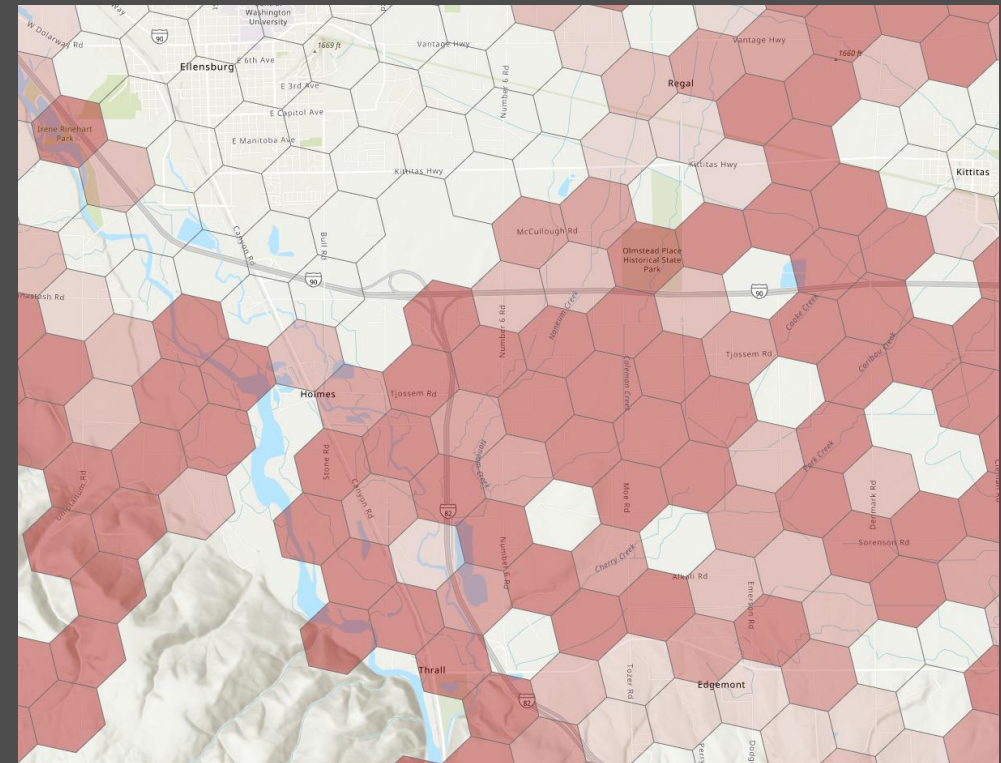


That sounds like a lot of work to get.

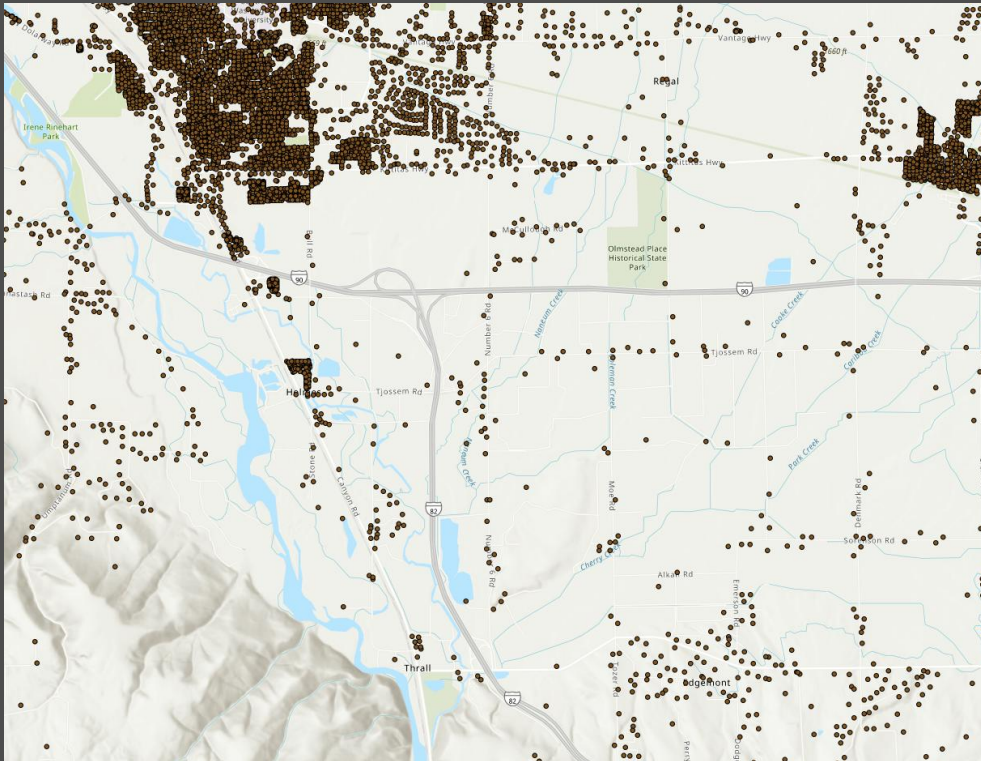
WSU

Pat Ryan's Living Atlas

- Fantastic data set from ESRI
- Has State, County, Tract, Block Group, Block, and H3R8 levels of geography
- [The Broadband Data Collection and ArcGIS Living Atlas of the World](#)
- [FCC Broadband Data Collection December 2023](#)



Site Address Point



- SAP Layer from the OCIO office
- State level dataset
- Pretty good
- Missing some counties
 - Looking at you Stevens, Lewis and Cowlitz
- Shout out to Columbia

WSU



REMIX!

- Using the Site Address Point layer you can create a very accurate understanding of availability
- Great tool for creating service areas, more on this in a bit

WSU

Other Datasets



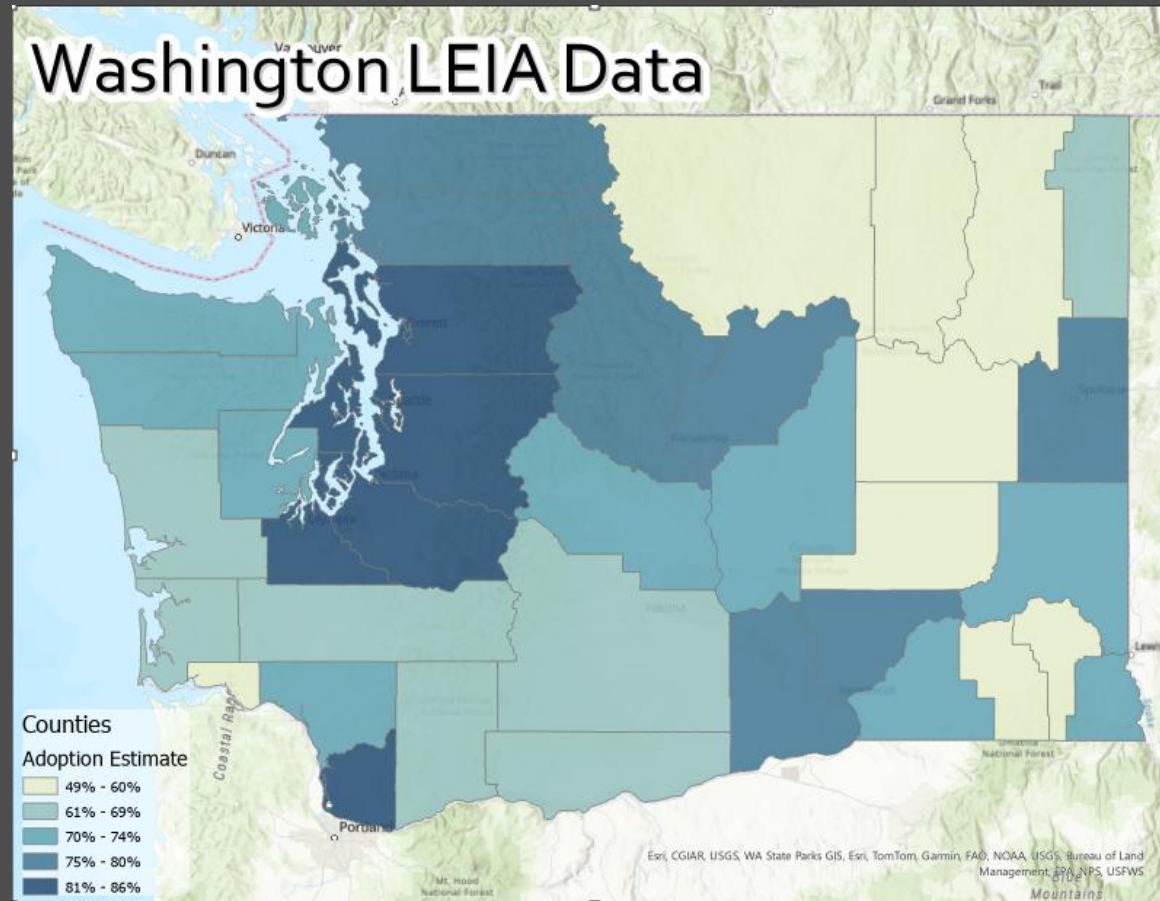
WSU

Local Estimates of Internet Adoption (LEIA)

- Census Bureau Experimental Dataset
- [2022 Local Estimates of Internet Adoption \(LEIA\) \(census.gov\)](#)

WSU

Local Estimates of Internet Adoption (LEIA)



Local Estimates of Internet Adoption (LEIA)

- US Census is currently seeking feedback on this data.

WSU

But what else could I do?



Lots of stuff!

WSSU

But what else could I do?



- Provider footprint analysis
- Growth area analysis
- Public engagement
- Many other ideas!

WSU

But why should I care now?



I mean it is a cool dataset and all

WSSU

BEAD



You have 1.23 Billion reasons to care

WSSU

Definitions



- Unserved: Under 25mbit down/3mbit up
- Underserved: Under 100 down/20 up equal to or above 25 down/3 up
- Served: Equal to or faster than 100 down/20 up
- 95th percentile latency less than 100ms for all definitions
- Able to be installed within 10 days
- Technologies: Wireline or Licensed Fixed Wireless
- Washington specifics:
 - 25/3 for every resident by 2024
 - 1g/1g for every CAI by 2026
 - 150/150 for every resident by 2028
- What is a BSL?
 - A business or residential location in the United States at which mass-market fixed broadband Internet access service is, or can be, installed.

Washington Timeline



- Challenge Process: Mid April 2024
- Final project areas: Now?
- NOFO: Soon (Late October 2024)
- Round 1: November 2024
- Round 2: January 2025
- Round 3: March 2025
- Final Proposal: April 2025
- Projects Start: 2024 (Maybe)
- Projects Finish: 2028

WSU

Correcting the Map



- Won't help with Federal allocations
- Will help with allocations of state funding
- Could help with making accurate Project Areas – More on this in a minute
- Will help with follow on programs

WSU

Community Anchor Institutions

What are they?

WVSU

Community Anchor Institutions



- K-12 Schools public or private (FCC E-rate or National Center for Education Statistics [NCES] ID Required)
- Higher Education (NCES ID Required)
- Libraries (FCC E-rate or American Library Association [ALA] member libraries and their branches)
- Health care facilities that have a Centers for Medicare and Medicaid Services (CMS) identifier
 - Health clinic
 - Hospitals
 - Mental health facilities
 - In/Out-patient treatment centers

WSU

Community Anchor Institutions



- Government Organizations
 - Tribal government buildings (TBD after further conversations with tribes and further reviews of the Community Action Plans)
 - State
 - WorkSource
 - DSHS - Community Services Offices
 - Community corrections
 - County
 - County Courthouse
 - Local
 - City/town hall
- Public Housing Organizations
 - Public housing as determined by state Public Housing Agencies (PHA) and enumerated by the US Department of Housing and Urban Development (HUD).
 - Housing listed on the National Housing Preservation Database

WSU

Community Anchor Institutions



- Community Support Organizations that facilitate greater use of broadband service by vulnerable pop, including, but not limited to, low-income individuals, the unemployed, children, incarcerated and aged individuals.
 - Youth Centers
 - Senior Centers
 - Job Training Centers
- Public Safety
 - Fire houses
 - Emergency medical service stations
 - Police stations
 - 911 PSAPs
 - Detention centers and correctional facilities

WSU

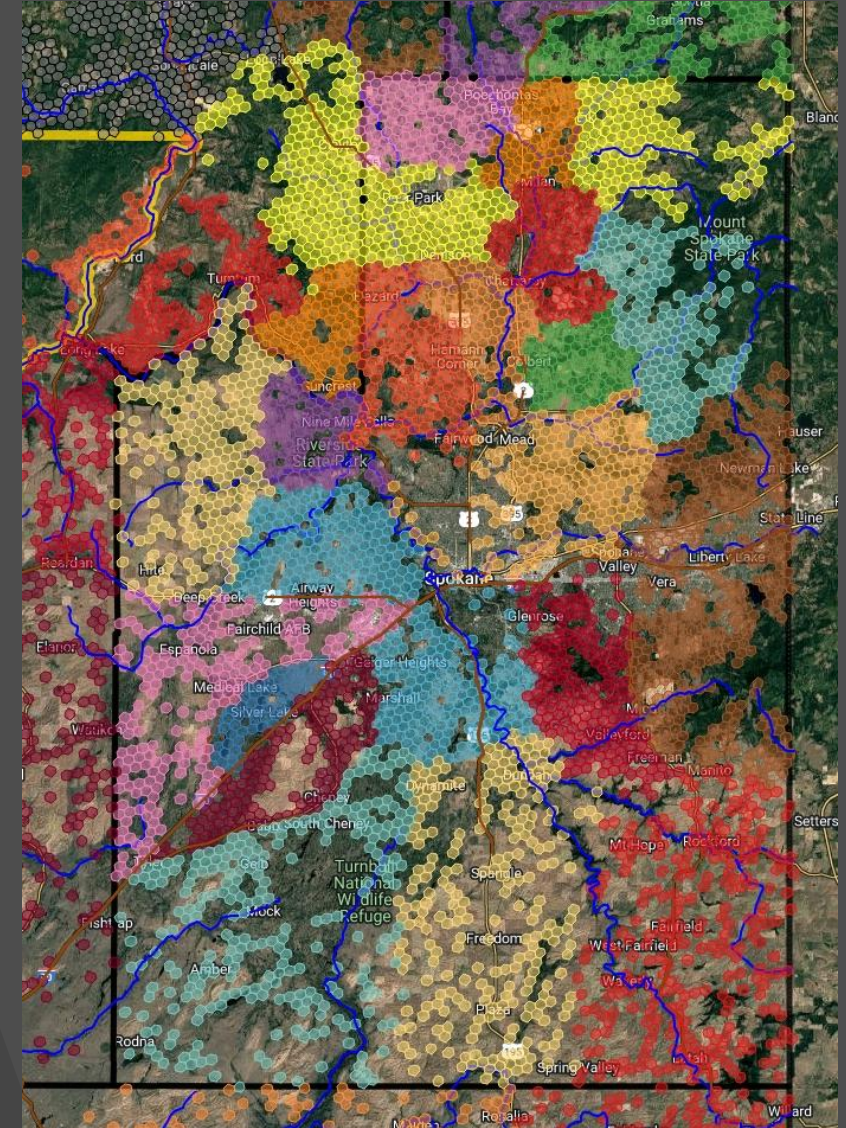
Project Areas



WSU

Project Areas

- Are going to be developed as a part of BEAD Proposals
- Currently being worked on by WSBO
- Changes are likely



Projects



- Every area will need to be applied for individually.
- Cohorts
 - Municipality
 - Infrastructure
 - Community
 - Everyone else!
- Each of these groups have an interest in the outcome of BEAD.

WSU

Beyond BEAD



There will be another

WSU

There will be another



And this data will guide the investments.

WSSU

Further Resources

- <https://broadband.wsu.edu>
 - Office Hours
 - Questions
- BDCToolbox – Toolbox for ArcGIS that I will be putting out by the end of Fall.
- Suggestions and questions are welcomed?

WSU

Thank you for your time today

WVSU